

Designing a Mixed Reality System for Exploring Genetic Mutation Data of Cancer Patients

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ABSTRACT

Increased availability of cancerous genomics mutation data provides researchers the opportunity to discover associations among genetic mutation patterns within the same organ as well as similar mutation patterns among different organs. However, the complexity, variety, and scale of multi-dimensional data involved in analyzing mutations across organs poses challenges for clinicians and researchers to draw such relationships. We present a prototype application that leverages multiple-coordinated views in mixed reality (MR) to enable investigations of genetic mutation patterns and the organs affected by cancer. We believe our prototype has the potential to enhance data and association discovery within and across different organs.

Index Terms: Information systems—Information systems applications—Decision support systems—Data analytics; Human-centered computing—Visualization—Visualization techniques; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed/augmented reality

1 INTRODUCTION

The Catalogue of Somatic Mutations in Cancer (COSMIC) is the world's largest source of expert curated somatic mutation information related to human cancers in the field, with data from over 26,000 peer reviewed papers [15]. Although complex, this data is invaluable for bioinformaticians, clinicians, and geneticists for predicting candidates for driver mutations in the genome, which in turn can improve patient diagnosis and prognosis [1]. However, the scale of data means that analysing it requires expertise, experience, and intuition. Various tools have attempted to aid in the exploration of genomics data, but most focus on creating a 2D visualization for the data. We present a novel method for displaying genomics data in immersive environments using multiple coordinated views with the goal of enhancing data association discovery within and across different organs.

2 RELATED WORK

2.1 Multiple-Coordinated Views

One strategy for increasing data discovery is through the use of multiple-coordinated views. This was highlighted by Scherr [13], who identified systems which leveraged both an “overview” and a “detail view”. In these systems the overview has a large portion of the dataset visible, while the detail view offers specific detail into a subset of the data. This provides the user with the needed context of the larger view while having the ability to see specific details. Scherr notes work both by Shimabukuro et al. [14] and Masui et al. [7] who applied “overview” and “detail view” into the domains of climate data exploration and city navigation, respectively. Work by Do Carmo et al. [4] also showed that users are able to coordinate between multiple data views in augmented reality easily

with usability studies showing “better and faster data comparisons and analysis” vs. desktop environments.

2.2 Genome Browsers

There has been ripe research into different interaction and visualization paradigms to aid with the traversal of dimensions in genomics data, particularly work by Nusrat et al. [11] and Nielsen et al. [10] note the challenges in comparing genomics data. In particular Nusrat et al. [11] notes multiple systems leverage multiple-coordinated views in 2D. However, no systems to our knowledge extend the use of such views in genomics data to an immersive 3D environment.

2.3 Biological Data Exploration Tools in VR/AR/MR

Various immersive environments have been researched for the use of data exploration in medicine. Examples include work Munzer et al. [9] and Hochman et al. [5] all of which show positive correlations between AR/VR and education in the field of medicine.

Researchers have also explored the potential applications of mixed and augmented reality for medical data exploration, with tools like BigTop, which focused on creating a Manhattan plot for genome-wide association studies (GWAS) datasets [16] and BioVR [17], which focused on modelling protein structure [17]. To the best of our knowledge, nothing like our prototype system exists for discovering associations among genetic mutation data.

3 PROTOTYPE DESIGN

3.1 Goal

Building on the related work, we aim to create a meaningful representations of the genomics data using multiple-coordinated views in MR. We believe this will improve data discovery and task completion speeds, as highlighted by Scherr [13], Shimabukuro et al. [14], Masui et al. [7], and Carmo et al. [4]. In particular, we created a mimetic overview, in the form of an avatar, to show the affected organs surrounded by a large immersive detailed view, in the form of a cylinder to show the genomics data. We believe this provides an improved environment for data discovery as supported by Cruz et al.'s [3] and Satriadi et al.'s [12] work in CAVE and VR immersive data visualizations.

3.2 Human Anatomy Avatar

Motivated by Moloney et al. [8], we include an overview mimetic representation of the affected tissues and sub-tissues in the form of a human anatomy avatar. This is important as Moloney et al. [8] note that mimetic representations of data can help align a user's natural experience with the virtual data representation, thereby providing affordances that abstract representations do not allow. We believe this has the potential to aid users in discovering correlations within and among cancers that affect different organs since the user can simultaneously visualize the genomics data for up to two tissues or sub-tissues.

The avatar consists of individual tissue actors in the Unreal level that have been labelled according to their naming in COSMIC. The actor has a HoloLens MRTK toggle button as a child, modified

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to be an interactive label. A user can examine different tissues or sub-tissues and spawn their genomics data visualization cylinder by selecting the corresponding label. Each tissue has a set of children sub-tissue labels, which become visible upon tissue selection.

To filter between cancers that affect one or both sexes, users can select the sex of the avatar using a toggle button located above it. This makes the labels of the affected tissues visible around the avatar, and allows the user to view areas of the model without the labels crowding their view. In a similar manner, sub-tissue labels only appear once a user has selected their respective tissue to avoid visually cluttering the user's field of view.

The A user can also scale and rotate the avatar to help with selection. Scaling and moving the avatar allows the user to position the avatar in the physical space as per the user's preference for comfort and ease.

3.3 Genomics Data Visualization Cylinder

Inspired by Manshaei et al.'s [6] 2D circular visualization, we display information about affected chromosomes and genes for each tissue or sub-tissue of patients in the database in a 3D cylindrical immersive visualization around the avatar. Representing patients as rings on the cylinder, rather than as concentric circles, allows us to have a consistent length for single chromosome or gene segments across patients. When the user taps on a tissue or sub-tissue label in the human anatomy avatar, a 'compressed' data visualization cylinder in the form of a discrete heatmap spawns around the avatar, showing a summation of all patient mutation data for that tissue at the chromosome level. Tapping on an affected chromosome segment allows the user to delve deeper into the data revealing a new cylinder highlighting that chromosome's affected genes. In a similar manner, a user can visualize mutations for all the patients for a specific chromosome or gene in the 'expanded' view. The user can transition from one format to another using a tapping gesture.

4 TECHNICAL OVERVIEW

Our system leverages the Microsoft HoloLens 2 headset as its MR head mounted display (HMD). The natively supported gaze and mid-air tap gestures along with developed composite gestures for more interactions aid in creating a sense of embodiment in the environment. This also offers the ability to further explore mid-air gestural input as an area of research that Besancon et al. notes as a method of input which is under-researched in the field [2].

We built our system in Unreal 4.27, as it provided the support for managing the IO and mesh generation efficiently through C++, as well as managing 3D assets. The mixed reality toolkit (MRTK v.2), provided by Microsoft, as an open-source Unreal toolkit for HoloLens 2 was used to implement our interactions. The MRTK supplements the lack of haptics in AR interactions through its included audio visual feedback. Additional data pre-processing was performed by a Python 3.8 API which serves the Unreal instance with formatted COSMIC data for use in the cylinder's mesh generation, as well as performing data clustering using TensorFlow. This helps offload computation from the machine running the visualization, to avoid lag.

5 CONCLUSION AND FUTURE WORK

In this paper, we explore the affordances of MR and multiple coordinated views for genomics data exploration. We reviewed previous work in immersive data discovery, genome browsers, and MR. Based on our findings we created an initial prototype system.

In the future we aim to test the system and interactions with user studies. One area we aim to explore further is mimetic interactions with the cylinder, such as opening and closing deeper layers by performing a movement like opening and closing a curtain, similar to Moloney et al. [8]. We also aim to add interactions that expose the underlying quantitative data for additional depth in exploration,

as well as implement tools for scaling the height of the cylinder up and down.

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